LESSON PLAN

AVIONICS AND ELECTRONIC WARFARE (13)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	19
Total No. of Lesson Plans	25	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Auxiliary starting devices: - Impulse coupling, booster coils ignition s/w and Demonstration of H.E.I.U operation.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Auxiliary starting devices: - Impulse coupling, booster coils ignition s/w and Demonstration of H.E.I.U operation. Safety precautions to be taken while working on high energy /voltage devices.
5. Specific Objectives	13.1.3.1 Knowledge on circuit protection devices, generators, actuators & engine starting system.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of H.E.I.U operation; Safety precautions for high energy/voltage devices; Instructor supervision during live firing
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of an impulse coupling in an engine starting system?
(b) Why are booster coils necessary in high-energy ignition systems?
(c) List two critical safety precautions when handling high-voltage ignition components.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Impulse coupling mechanism and operation • Function of booster coils in ignition • Ignition switch wiring and circuit protection • High Energy Ignition Unit (H.E.I.U) components • Safety protocols for high voltage devices	L & Ds	PPT, IB	15 min
(b)	• Detailed H.E.I.U operational sequence • Circuit protection devices and generators • Actuator functions in starting system • Live demonstration of H.E.I.U firing • Troubleshooting high energy ignition faults	L & Ds	PPT, PDF	20 min
(c)	• Review of safety precautions • Summary of auxiliary starting devices • Q&A on circuit protection • Linkage to next lesson on ice protection • Final safety checklist verification	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how the impulse coupling ensures proper spark timing during engine start.
(b) Describe the role of the H.E.I.U in generating high-energy sparks for ignition.

12. Home Assignment and Follow Up	Prepare a brief note on the pneumatic de-icing system principles to be discussed in the next lesson on Ice and rain protection systems.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Ice and rain protection systems, methods of protection, pneumatic de-icing system, electrical de-icing and anti-icing system.	Refer to the master syllabus reference list	LP 20

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min